

THE EFFECT OF AN ELECTROSTATIC FIELD ON PARTICLE ACCELERATION AT CYCLOTRON RESONANCE

D. C. SCHRAM and G. P. BEUKEMA*

*Association Euratom-FOM, FOM-Instituut voor Plasma-Fysica
Rijnhuizen, Jutphaas, Nederland*

Received 11 September 1968

Synopsis

The effect of an electrostatic field parallel to a static magnetic field on the cyclotron resonance acceleration of electrons is studied. Exact relativistic calculations show that for electrostatic field strengths below a certain limit, which depends on the field strength of the e.m. wave and the index of refraction, a steady average energy gain occurs. At this limit the energy of the particle increases steadily without any oscillation, while above the limit a stepwise energy gain occurs. Sufficiently far above, an analytical nonrelativistic approximation yields a simple expression for the energy gain of electrons with an initially isotropic transverse velocity distribution.

1. *Introduction.* At cyclotron resonance an electron would gain an infinite energy if the resonance condition

$$r \equiv \frac{\omega - kv_z - \Omega(t)}{\omega} = 0$$

were sustained for an infinitely long time. We shall consider the collisionless case and uniform electromagnetic (frequency ω , wavenumber k) and static magnetic fields. The first and second order Doppler effects, *i.e.* the variations of the axial velocity component v_z and the relativistic variation of the mass of the electron, will make kv_z and $\Omega(t) = eB_0/\{m(t)\omega\}$, and, therefore, also r time-dependent quantities^{2,3}). Consequently, an energy limitation occurs, except for the synchronous case^{1,2}): $v_p = c$ and $r(0) = 0$, where the two Doppler effects cancel each other. Generally there exists an implicit relationship between both variations³):

$$\frac{d\gamma}{dt} = \frac{v_p}{c} \frac{d}{dt} \left(\gamma \frac{v_z}{c} \right), \quad (1)$$

* Present address: Fysisch Laboratorium der Rijksuniversiteit, Utrecht, Nederland.

where γ is the normalized total energy of the particle:

$$\gamma = \frac{\mathcal{E}}{mc^2} = \left(1 - \frac{\mathbf{v} \cdot \mathbf{v}}{c^2}\right)^{-1/2} = \frac{m(t)}{m}. \quad (2)$$

Here $m(t)$ is the time-dependent apparent mass and m is the rest mass of the particle.

If an electrostatic field, $\mathbf{E}_0$, directed parallel to the static magnetic field, $\mathbf{B}_0$, is present, the relationship (1) is changed. The axial acceleration is not only caused by the high frequency force, $-e\mathbf{v}_\perp \times \mathbf{B}_\perp^\sim$, but also by the electrostatic force, $-e\mathbf{E}_0$. The electrostatic force can change the axial velocity in such a way, that the resonance condition will be fulfilled for a certain period of time. We shall call this effect pulling the electron through resonance. During the passage through resonance the electron energy may increase. In the following sections this increase will be calculated in an exact numerical treatment as well as analytically in a nonrelativistic approximation.

Similar phenomena occur, without an electrostatic field, if a particle passes a resonant zone in an inhomogeneous static magnetic field^{4, 5)}, or if the magnetic field $\mathbf{B}_0$ is varied in time⁶⁾. This is quite understandable, as in all cases the electron is pulled through resonance by external means.

2. Exact numerical treatment. In a recent study³⁾, describing the influence of the first and second order Doppler effects on the motion of an electron, an electrostatic field term has already been included indicatively in the formalism. In that treatment, which we shall briefly resume here, it appeared to be of advantage to write the equations in a coordinate system $(\mathbf{i}_\xi, \mathbf{i}_\eta, \mathbf{i}_z)$ which rotates and travels with the right-handed circular polarized TEM wave:

$$\begin{aligned} \mathbf{i}_\xi &= \mathbf{i}_x \cos(\omega t - kz) + \mathbf{i}_y \sin(\omega t - kz), \\ \mathbf{i}_\eta &= -\mathbf{i}_x \sin(\omega t - kz) + \mathbf{i}_y \cos(\omega t - kz), \\ \mathbf{i}_z &= \mathbf{i}_z. \end{aligned} \quad (3)$$

In this frame the e.m. wave and static fields are (cf. fig. 1):

$$\begin{aligned} \mathbf{E}^\sim &= -\hat{E}\mathbf{i}_\eta; & \mathbf{B}^\sim &= \frac{\hat{E}}{v_p} \mathbf{i}_\xi \quad \text{e.m. wave;} \\ \mathbf{E}_0 &= E_0\mathbf{i}_z = -\frac{\partial\phi(z)}{\partial z} \mathbf{i}_z; & \mathbf{B}_0 &= B_0\mathbf{i}_z \quad \text{static fields.} \end{aligned} \quad (4)$$

The velocity components in this frame are:

$$v_\xi = \mathbf{v} \cdot \mathbf{i}_\xi, \quad v_\eta = \mathbf{v} \cdot \mathbf{i}_\eta, \quad v_z = \mathbf{v} \cdot \mathbf{i}_z, \quad (5)$$

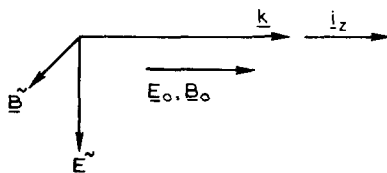


Fig. 1. The orientation of the fields.

and normalized momenta are:

$$p_{\xi} \equiv \gamma \frac{v_{\xi}}{c}; \quad p_{\eta} \equiv \gamma \frac{v_z}{c}. \quad (6)$$

A set of three coupled differential equations derived from the equation of motion describes the behaviour of an electron in the given fields:

$$\frac{d\gamma}{d\tau} = \frac{v_p}{c} \frac{d}{d\tau} \left(\gamma \frac{v_z}{c} \right) - \left(1 - \frac{v_z}{v_p} \right) \Phi', \quad (7)$$

$$\frac{d}{d\tau} \left\{ \frac{1}{1 - b/\gamma - v_z/v_p} \frac{dp_{\xi}}{d\tau} \right\} = - \left(1 - \frac{b}{\gamma} - \frac{v_z}{v_p} \right) p_{\xi} + \left(1 - \frac{v_z}{v_p} \right) g, \quad (8)$$

$$\gamma \frac{v_p}{c} \frac{d}{d\tau} \left(\gamma \frac{v_z}{c} \right) - \gamma \Phi' = \frac{g}{1 - b/\gamma - v_z/v_p} \frac{dp_{\xi}}{d\tau} = g p_{\eta}. \quad (9)$$

Here the following definitions have been used:

$$\left. \begin{aligned} b &= \frac{\Omega}{\omega} = \frac{eB_0}{m\omega} && \text{static magnetic field parameter}^*, \\ g &= \frac{e\hat{E}}{m\omega c} && \text{wave electric field parameter,} \\ \Phi' &= \frac{d}{d(kz)} \frac{e\phi}{mc^2} = \frac{eE_0 v_p}{m\omega c^2} && \text{electrostatic field parameter,} \end{aligned} \right\} \quad (10)$$

$$\tau = \omega t \quad \text{reduced time.} \quad (11)$$

There are three independent initial conditions:

$$\gamma(t=0) = \gamma_0, \quad p_{\xi}(t=0) = p_{\xi 0} = \gamma_0 \frac{v_{x0}}{c}, \quad v_z(t=0) = v_{z0}. \quad (12)$$

The set of equations (7), (8) and (9) has been programmed on an analog computer. The electrostatic field is assumed to be spatially uniform. For

* Note, that while $b = eB_0/(m\omega) = \Omega/\omega$ is the reduced cyclotron frequency related to the rest mass, $b/\gamma = eB_0/\{m(t)\omega\} = \Omega(t)/\omega$ is the reduced cyclotron frequency related to the time-varying apparent mass of the particle.

several values of the parameters and initial values the time- and z -dependences of the particle energy have been calculated. For a qualitative description of the numerical results some features of the solution for a zero electrostatic field will be of interest.

If there is no electrostatic field, $\Phi' = 0$, the set of equations (7), (8) and (9) can be reduced to one differential equation which has been solved in refs. 2 and 3. The energy oscillates as a function of time and of z (curves labelled -0- in figs. 2a and 2b) in a way related to the time dependence of the resonance function r :

$$r = 1 - \frac{v_z}{v_p} - \frac{b}{\gamma}. \quad (13)$$

As can be seen from eq. (1), which can be rewritten as follows³):

$$\frac{1}{b/\gamma} \frac{d(b/\gamma)}{d\tau} = \frac{1}{\left[1 - \frac{v_z v_p}{c^2}\right]} \frac{d}{d\tau} \left(1 - \frac{v_z v_p}{c^2}\right); \quad (14)$$

the relative importance of the two relativistic effects, schematically indicated as $\Delta(b/\gamma)$ and $\Delta(v_z/v_p)_{\text{hf}}$, depends on v_p/c . In a slow wave, $v_p < c$, the effect of the variation of the axial velocity, $\Delta(v_z/v_p)_{\text{hf}}$, is dominant, while in a fast wave, $v_p > c$, the relativistic mass variation, $\Delta(b/\gamma)$, is more important. If v_p differs not too much from c , both effects are of the same order of magnitude. For $v_p = c$, they exactly cancel each other¹): Davydovskii-effect (curve labelled -0- in fig. 2c). Therefore, then both variations $\Delta(b/\gamma)$ and $\Delta(v_z/v_p)_{\text{hf}}$ can be neglected as far as the calculation of the time behaviour of the transverse energy is concerned. However, the variation of the axial velocity, which is in fact due to the high frequency force in the axial direction, may not be neglected if the z -dependence of the energy is calculated.

For a nonzero electrostatic field eq. (7) is the extended form of eq. (1). There exists an additional variation of the axial velocity, $\Delta(v_z/v_p)_{\text{es}}$, due to the action of the electrostatic field. However, the arguments presented above concerning the combined effects of the variations of relativistic nature, $\Delta(b/\gamma)$ and $\Delta(v_z/v_p)_{\text{hf}}$, remain valid. Hence, in case $\Delta(v_z/v_p)_{\text{hf}}$ can be neglected with respect to $\Delta(v_z/v_p)_{\text{es}}$, and if v_p differs not too much from c , also $\Delta(b/\gamma)$ must be neglected in r . This has been done in the nonrelativistic approximation in section 3.

Some indicative results of the exact relativistic treatment will be discussed below. In all cases studied, the wave-field parameter is chosen to be $g/\gamma_0 = 2 \times 10^{-3}$ and the magnetic-field parameter $b/\gamma_0 = 1^*$. Three cases

* For $b/\gamma_0 \neq 1$, the qualitative behaviour is similar.

are studied:

- zero initial velocity: $v_{z0} = 0, v_{\perp 0} = 0$ ($\mathbf{v}_0 = \mathbf{0}$);
- zero initial transverse velocity: $v_{z0} \neq 0, v_{\perp 0} = 0$;
- nonzero initial velocity: $v_{z0} \neq 0, v_{\perp 0} \neq 0$.

a. $v_{\perp 0} = 0, v_{z0} = 0$.

The particle starts at resonance: $\rho \equiv r(0) = 1 - b/\gamma_0 - v_{z0}/v_p = 0$. In figs. 2a, 2b and 2c the energy variation is shown as function of $\tau = \omega t$ and of kz for three values of the refractive index: $c/v_p < 1, c/v_p > 1$, and $c/v_p = 1$, and for several values of Φ' . The case $\Phi' = 0$ is included, showing (dashed lines), according to ref. 3, a steady oscillatory behaviour of the energy. For a nonzero Φ' two types of energy gain occur: either a steadily increasing oscillatory energy variation (curves labelled 0.1, 1, 1.5, 1.65 in fig. 2a, and $-0.1, -0.5, -0.97$ in fig. 2b), or a step in energy followed by a decaying energy oscillation (other curves in fig. 2a, 2b and 2c). The existence of different types of energy gain can be explained qualitatively, *e.g.* for the fast wave $v_p > c$ as follows: In a fast wave, $\Delta(b/\gamma) > \Delta(v_z/v_p)_{\text{hf}}$. A small $\Phi' > 0$ will somewhat enlarge $\Delta(v_z/v_p)_{\text{total}} = \Delta(v_z/v_p)_{\text{hf}} + \Delta(v_z/v_p)_{\text{es}}$: during every oscillation the resonance function r gets closer to zero. The

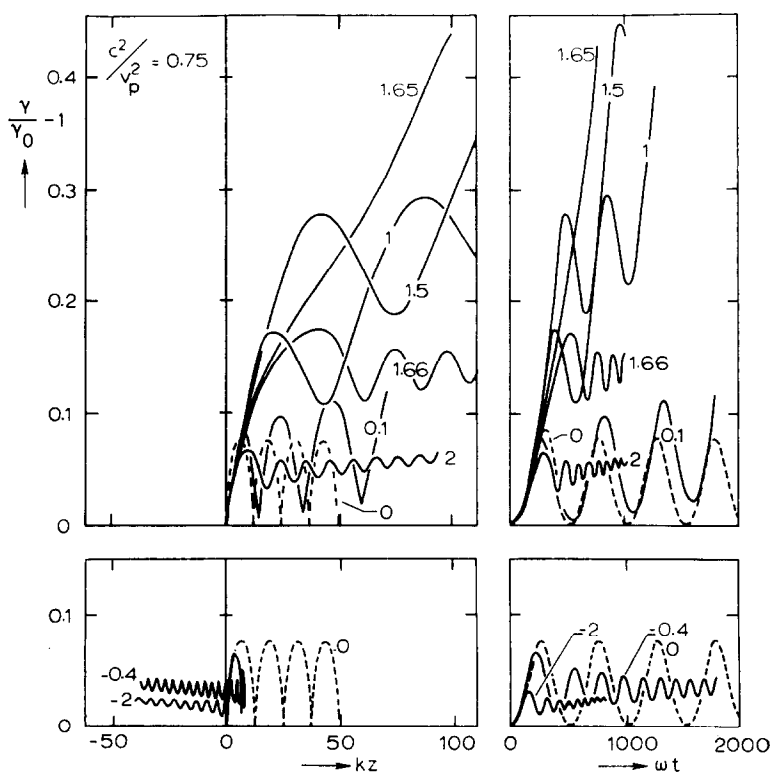


Fig. 2a

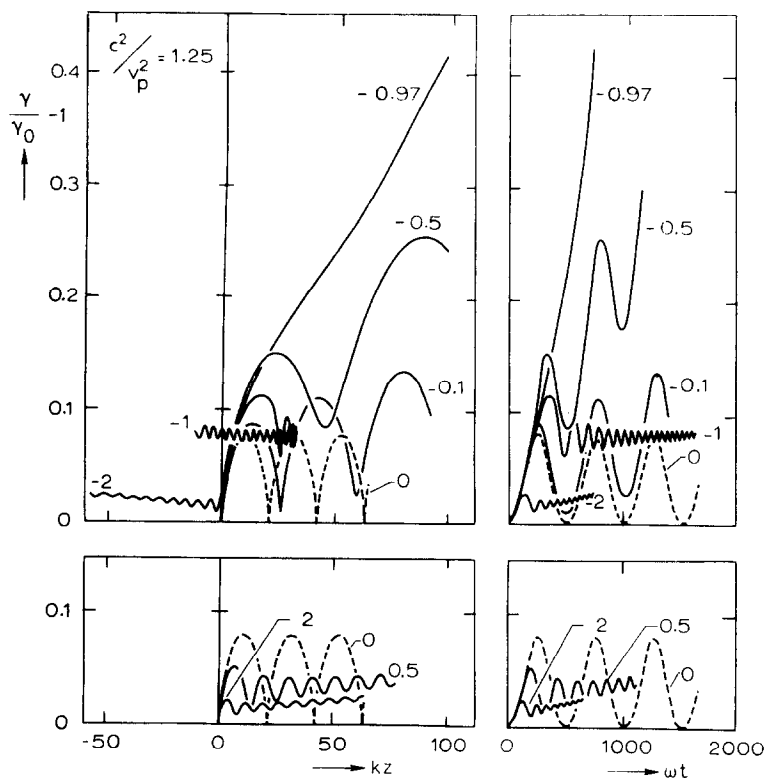


Fig. 2b

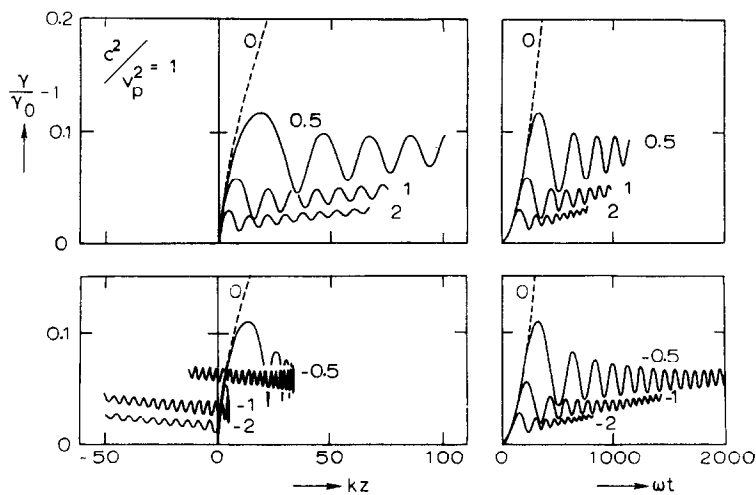


Fig. 2c

Fig. 2. Energy increase as function of displacement and time for $v_0=0$, $g/\gamma_0=2 \times 10^{-3}$, $b/\gamma_0=1$, for three values of $c^2/v_p^2=0.75, 1.25$ and 1 (figs. a, b and c), and with $10^4 \Phi'/\gamma_0$ as parameter.

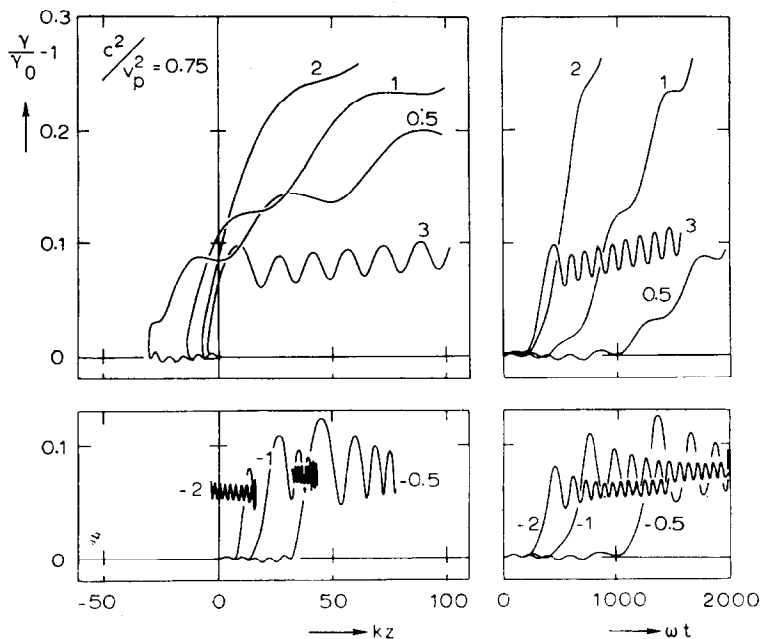


Fig. 3a

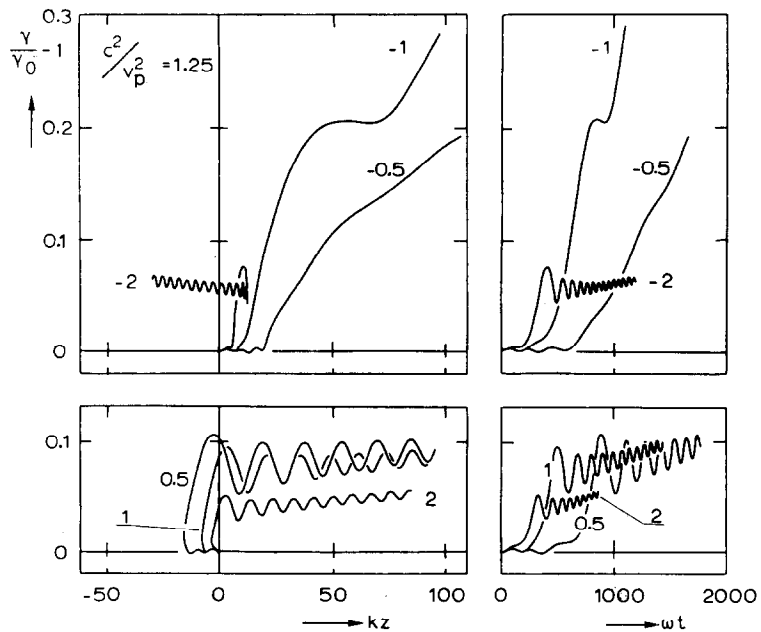


Fig. 3b

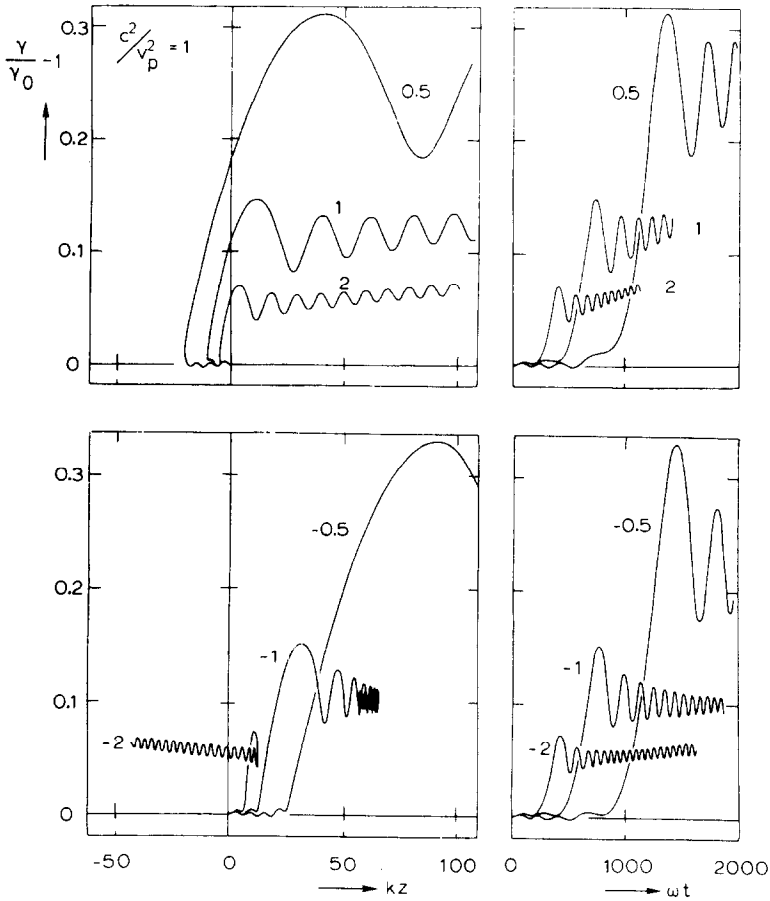


Fig. 3c

Fig. 3. Energy increase as function of displacement and time for $v_{\perp 0} = 0$, $v_{z0}/v_p = \pm 0.05$, $g/\gamma_0 = 2 \times 10^{-3}$, $b/\gamma_0 = 1$, for three values of $c^2/v_p^2 = 0.75, 1.25$ and 1 (figs. a, b and c), and with $10^4 \Phi'/\gamma_0$ as parameter.

energy gain will be higher than without an electrostatic field. A large positive Φ' will cause a large $\Delta(v_z/v_p)_{es}$, which rapidly overcompensates the difference between $\Delta(b/\gamma)$ and $\Delta(v_z/v_p)_{hr}$: during the first oscillation the electron is pulled out of resonance. At the transition value, Φ'_{res} , $\Delta(v_z/v_p)_{es}$ compensates the difference between $\Delta(v_z/v_p)_{hr}$ and $\Delta(b/\gamma)$ for a long time and the energy increases steadily (curve labelled 1.65 in fig. 2a). From estimating the effects of $\Delta(b/\gamma)$, $\Delta(v_z/v_p)_{hr}$ and $\Delta(v_z/v_p)_{es}$, we find that Φ'_{res} fulfils the condition:

$$\frac{c^2}{v_p^2} \Phi'_{res} = C \{g^2(1 - c^2/v_p^2)\}^{2/3} \frac{(1 - c^2/v_p^2)}{|1 - c^2/v_p^2|}, \quad (15)$$

where the constant C is numerically found to be $C \approx 1.2$. A negative Φ' will cause a $\Delta(v_z/v_p)_{\text{es}} < 0$, and will pull the electron quickly out of resonance. Hence, only the stepwise energy gain will occur.

For a slow wave, $\Delta(b/\gamma) < \Delta(v_z/v_p)_{\text{hr}}$. Consequently, the increasing type of energy occurs for negative values of Φ' . Again there is a resonant transition value Φ'_{res} which satisfies condition (15), see fig. 2b.

In free space, $v_p = c$, neither a positive nor a negative Φ' can cause the oscillatory increasing type of energy gain. For all values of Φ' the stepwise energy gain occurs. In other words, the resonant value of Φ' is $\Phi'_{\text{res}} = 0$, which satisfies condition (15). This result is identical to the Davydovskii effect¹⁾. It is also clear that the transverse energy behaviour as function of time for positive and negative values of Φ' are similar (fig. 2c) and can be approximated in the nonrelativistic treatment of section 3. Note that in the z -dependence of the energy the high frequency force is apparent.

b. $v_{z0} \neq 0$, $v_{\perp 0} = 0$.

Here the particle starts out of resonance: $\rho = -v_{z0}/v_p \neq 0$. Only that electrostatic field which is directed such that it takes the particle to resonance will be considered; any Φ' of opposite sign will only slightly modify the behaviour: if $v_{z0} > 0$, then $\Phi' < 0$ and if $v_{z0} < 0$, then $\Phi' > 0$. Interactions with a fast wave, with a slow wave, and in free space are shown in figs. 3a, 3b and 3c. The increasing and resonant types of energy gain occur again in a fast wave for positive values of Φ' and in a slow wave for negative values of Φ' . The resonant value Φ'_{res} follows roughly relation (15).

c. $v_{z0} \neq 0$, $v_{\perp 0} \neq 0$.

From the curves shown in fig. 4 it appears that the energy gain depends on $v_{\perp 0}$. This dependence is only negligible for $v_{\perp 0}/c \ll g/\sqrt{|\Phi'|}$. Because of the mathematical complexity of the problem it is cumbersome to obtain information about the energy increase of a population with an initially

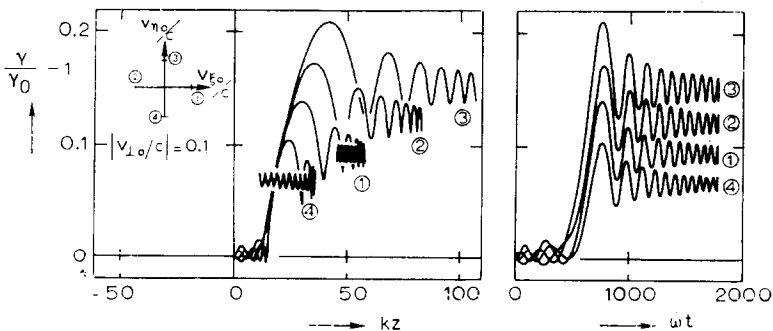


Fig. 4. Energy increase as function of displacement and time for $|v_{\perp 0}/c| = 0.1$, $v_{z0}/v_p = 0.05$, $g/\gamma_0 = 2 \times 10^{-3}$, $b/\gamma_0 = 1$, $\Phi'/\gamma_0 = -10^{-4}$, $c^2/v_p^2 = 1$, and four phases of $v_{\perp 0}$.

isotropic transverse velocity distribution. In the nonrelativistic approximation an analytical treatment is possible.

3. *The nonrelativistic approximation.* Again Φ' is assumed to be spatially uniform. A comparison of the combined effects of relativistic nature with the effect of the electrostatic field yields that a nonrelativistic approximation is valid for:

$$g^{4/3}(1 - (c/v_p)^2)^{2/3} \ll \frac{c^2}{v_p^2} |\Phi'|. * \quad (16)$$

This approximation is characterized by the simultaneous omission of $\Delta(b/\gamma)$ and $\Delta(v_z/v_p)_{\text{nr}}$. For v_p close to c the approximation remains valid even for small values of Φ' . This is due to the earlier mentioned effect that $\Delta(v_z/v_p)_{\text{nr}} = -\Delta(b/\gamma)$ when $v_p = c$. As stated in section 2, $\Delta(v_z/v_p)_{\text{nr}}$ may not be neglected in the calculation of the axial energy. The nonrelativistic approximation yields a simple solution of the transverse energy $\mathcal{E}_\perp$ as function of time. The transverse and longitudinal components of the equation of motion are decoupled:

$$r = 1 - b/\gamma - v_z/v_p \approx 1 - b - v_{z0}/v_p - a\tau, \quad (17)$$

where

$$a \equiv \frac{c^2}{v_p^2} \Phi'. \quad (18)$$

Substitution of eq. (17) in the transverse components of the equation of motion gives:

$$\frac{d}{d\tau} \left\{ \frac{1}{\rho - a\tau} \frac{dv_\xi/c}{d\tau} \right\} = -(\rho - a\tau) \frac{v_\xi}{c} + (\rho + b - a\tau) g. \quad (19)$$

This result may easily be checked by substituting eq. (17) in eq. (7). In terms of reduced quantities:

$$\psi_\xi = \frac{(v_\xi - g) a}{bg\sqrt{|a|}}, \quad \psi_\eta = \frac{v_\eta}{bg} \sqrt{|a|}; \quad (20)$$

$$T = (\tau - \rho/a) \sqrt{|a|}, \quad (21)$$

the differential equation (19) becomes:

$$\frac{d^2\psi_\xi}{dT^2} - \frac{1}{T} \frac{d\psi_\xi}{dT} + T^2\psi_\xi = -T. \quad (22)$$

The quantity ψ_η can be expressed in ψ_ξ as follows:

$$\psi_\eta = -\frac{1}{T} \frac{d\psi_\xi}{dT}. \quad (23)$$

* Note the similarity to eq. (15).

The solution of eq. (22) is ⁷⁾:

$$\begin{aligned}\psi_{\xi}(T) = & \cos \frac{1}{2}T^2 \int_{T_0}^T \sin \frac{1}{2}\theta^2 d\theta - \sin \frac{1}{2}T^2 \int_{T_0}^T \cos \frac{1}{2}\theta^2 d\theta + \\ & + C_1 \sin \frac{1}{2}T^2 + C_2 \cos \frac{1}{2}T^2.\end{aligned}\quad (24)$$

Substitution of the solution of $\psi_{\xi}(T)$ in equation (23) gives:

$$\begin{aligned}\psi_{\eta}(T) = & \sin \frac{1}{2}T^2 \int_{T_0}^T \sin \frac{1}{2}\theta^2 d\theta + \cos \frac{1}{2}T^2 \int_{T_0}^T \cos \frac{1}{2}\theta^2 d\theta + \\ & - C_1 \cos \frac{1}{2}T^2 + C_2 \sin \frac{1}{2}T^2.\end{aligned}\quad (25)$$

The integration constants C_1 and C_2 follow from the initial conditions:

$$\begin{aligned}C_1 = & \psi_{\xi 0} \sin \frac{1}{2}T_0^2 - \psi_{\eta 0} \cos \frac{1}{2}T_0^2, \\ C_2 = & \psi_{\xi 0} \cos \frac{1}{2}T_0^2 + \psi_{\eta 0} \sin \frac{1}{2}T_0^2,\end{aligned}\quad (26)$$

where an index 0 signifies the initial value of the quantity at $t(0) = 0$. The integrals occurring in the solution for $\psi_{\xi}(T)$ and $\psi_{\eta}(T)$ can be expressed in the Fresnel integrals:

$$\left. \begin{aligned}C(T^2) = & \pm \sqrt{\frac{2}{\pi}} \int_0^T \cos \theta^2 d\theta \\ S(T^2) = & \pm \sqrt{\frac{2}{\pi}} \int_0^T \sin \theta^2 d\theta\end{aligned} \right\} \text{ for } T = \pm |T|.\quad (27)$$

It has already been remarked before that the electrostatic field has only a significant influence if it pulls the electron through resonance. This means mathematically that T must change sign from $T_0 = -\rho/a < 0$ to $T > 0$. $T = 0$ indicates the moment in time that the particle passes through the resonance.

With the different definitions (27) of the Fresnel integrals at $T > 0$ and at $T_0 < 0$, eqs. (24) and (25) become:

$$\begin{aligned}\psi_{\xi} = & \psi_{\xi 0} \cos \frac{1}{2}(T^2 - T_0^2) - \psi_{\eta 0} \sin \frac{1}{2}(T^2 - T_0^2) + \\ & + \sqrt{\pi} [(S + S_0) \cos \frac{1}{2}T^2 - (C + C_0) \sin \frac{1}{2}T^2], \\ \psi_{\eta} = & \psi_{\xi 0} \sin \frac{1}{2}(T^2 - T_0^2) + \psi_{\eta 0} \cos \frac{1}{2}(T^2 - T_0^2) + \\ & + \sqrt{\pi} [(S + S_0) \sin \frac{1}{2}T^2 + (C + C_0) \cos \frac{1}{2}T^2],\end{aligned}\quad (28)$$

where

$$S \equiv S(\frac{1}{2}T^2), \quad S_0 \equiv S(\frac{1}{2}T_0^2), \quad C \equiv C(\frac{1}{2}T^2), \quad C_0 \equiv C(\frac{1}{2}T_0^2).$$

The normalized kinetic transverse energy increase is:

$$\begin{aligned}
 \frac{\mathcal{E}_\perp - \mathcal{E}_{\perp 0}}{mc^2} = & \frac{\pi b^2 g^2}{2|a|} [(S + S_0)^2 + (C + C_0)^2] + \\
 & + bg \sqrt{\frac{\pi}{|a|}} \left[(S + S_0) \left(\frac{a}{|a|} \frac{v_{\xi 0}}{c} \cos \frac{1}{2} T_0^2 + \frac{v_{\eta 0}}{c} \sin \frac{1}{2} T_0^2 \right) + \right. \\
 & - (C + C_0) \left(\frac{a}{|a|} \frac{v_{\xi 0}}{c} \sin \frac{1}{2} T_0^2 - \frac{v_{\eta 0}}{c} \cos \frac{1}{2} T_0^2 \right) \left. \right] + \\
 & + \frac{a}{|a|} b g^2 \sqrt{\frac{\pi}{|a|}} \{ (S + S_0) (\cos \frac{1}{2} T^2 - \cos \frac{1}{2} T_0^2) + \\
 & - (C + C_0) (\sin \frac{1}{2} T^2 - \sin \frac{1}{2} T_0^2) \} + \\
 & + g \left[\frac{v_{\xi 0}}{c} \{ \cos \frac{1}{2} (T^2 - T_0^2) - 1 \} - \frac{a}{|a|} \frac{v_{\eta 0}}{c} \sin \frac{1}{2} (T^2 - T_0^2) \right] + \\
 & + g^2 [1 - \cos \frac{1}{2} (T^2 - T_0^2)]. \tag{29}
 \end{aligned}$$

The energy behaviour as function of time is similar to the steptype in the relativistic treatment; compare the dotted and full lines in fig. 5. For $c^2/v_p^2 = 1$, both treatments give nearly equal results even for small values of Φ'/γ_0 (figs. 5b and 5d). For $c^2/v_p^2 \neq 1$, there is a small discrepancy; however, the values chosen for Φ'/γ_0 are just above the values given by inequality (16) (figs. 5a and 5c). As in general $|a| \ll 1$, only the first two terms of the right-hand side of eq. (29) have to be considered. For an initial transverse velocity $|v_{\perp 0}| \ll g/\sqrt{|a|}$ only the first term is of importance. If S , S_0 , C and C_0 are replaced by 0.5, which means that both $|T_0|$ and $|T|$ are not too small, the energy gain of an electron with small initial transverse velocity is:

$$\frac{\mathcal{E}_\perp - \mathcal{E}_{\perp 0}}{mc^2} = \pi \frac{b^2 g^2}{|a|}. \tag{30}$$

If $v_{\perp 0}$ may not be neglected, the energy gain, or even loss, depends not only on $|v_{\perp 0}|$, but also on the phase relationship between $\mathbf{E}$ and $\mathbf{v}_{\perp 0}$ (see also fig. 5). For an electron population with constant $|v_{\perp 0}|$ and isotropic distributed initial phases between $\mathbf{E}$ and $\mathbf{v}_{\perp 0}$, the energy gain per electron is approximately:

$$\frac{\mathcal{E}_\perp - \mathcal{E}_{\perp 0}}{mc^2} = \frac{\pi b^2 g^2}{2|a|} [(S + S_0)^2 + (C + C_0)^2]. \tag{31}$$

As this result does not depend on the magnitude of $v_{\perp 0}$, it is also valid for a population with any isotropic transverse distribution function. Replacing

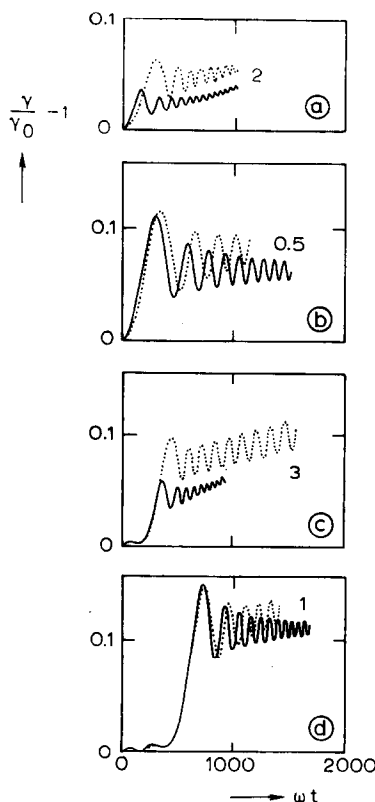


Fig. 5. Comparison of energy increases as function of time according to relativistic (dotted line) and nonrelativistic treatments. In all figures $g/\gamma_0 = 2 \times 10^{-3}$, $b/\gamma_0 = 1$, $v_{\perp 0} = 0$. For fig. a, b: $v_{z0} = 0$ and for c, d: $v_{z0}/v_p = 0.05$; for fig. a, c: $c^2/v_p^2 = 0.75$ and for b, d: $c^2/v_p^2 = 1$. $10^4 \Phi'/\gamma_0$ is parameter.

again S , S_0 , C and C_0 by 0.5 the energy gain per electron reduces to:

$$\frac{\mathcal{E}_{\perp} - \mathcal{E}_{\perp 0}}{mc^2} = \pi \frac{b^2 g^2}{|a|}. \quad (32)$$

Acknowledgements. The authors are indebted to Professor C. M. Braams and to Dr. L. Th. M. Ornstein for many helpful discussions and suggestions. This work was performed as part of the research program of the association agreement of Euratom and the "Stichting voor Fundamenteel Onderzoek der Materie" (FOM) with financial support from the "Nederlandse Organisatie voor Zuiver-Wetenschappelijk Onderzoek" (ZWO) and Euratom.

REFERENCES

- 1) Davydovskii, V. Ya., Soviet Physics JETP **16** (1963) 629.
- 2) Roberts, C. S. and Buchsbaum, S. J., Phys. Rev. **135** (1964) A381.
- 3) Schram, D. C., Physica **37** (1967) 617.
- 4) Seidl, M., Plasma Physics, J. nuclear Energy (part C) **6** (1964) 597.
- 5) Tuma, D. T. and Lichtenberg, A. J., Plasma Physics, J. nuclear Energy (part C) **9** (1967) 87.
- 6) Ornstein, L. Th. M., Private Communication.
- 7) Kamke, E., Differentialgleichungen I, Akademische Verlagsgesellschaft, Geest and Portig K.G. (Leipzig, 1959).